Feature Flag Management System for Tatum Bank

1. Introduction

We are proposing a Feature Flag Management System for Tatum Bank. The system will allow administrators to control the availability of features in the bank's application by turning them on or off without requiring a new application release every time a setting changes.

For example, Tatum Bank may introduce a betting feature but decide not to make it available to everyone immediately. Administrators could use the feature flag system to control its availability while the feature is being tested or gradually introduced to users.

The betting feature is only an example. The system is designed to manage different features within Tatum Bank's applications.

2. Problem Statement

When an application introduces a new feature, developers may need to change application code and release an update to control when that feature becomes available.

This can make it difficult to introduce features gradually, limit availability to selected users, or disable a feature when necessary.

The proposed system aims to solve this problem by separating feature availability settings from the code that implements each feature. This allows administrators to manage feature availability through settings rather than requiring a new release for every change.

3. Objectives

The objectives of the system are to:

- Allow administrators to register and manage application features.
- Allow features to be enabled or disabled without requiring a new application release.
- Support different settings for different environments, such as Development, Staging, and Production.
- Allow administrators to control which users can access a feature.
- Provide a consistent way for applications to check whether a feature should be available.

4. Functional Requirements

The functional requirements describe what the system must be able to do.

FR1 — Application Management: The system must allow administrators to register applications whose features will be managed.

FR2 — Feature Management: The system must allow administrators to register features under an application.

FR3 — Global Feature Control: The system must allow administrators to enable or disable a feature for a particular environment.

FR4 — User Targeting: The system must allow administrators to restrict a feature to selected users when needed.

FR5 — Environment Management: The system must support separate feature settings for Development, Staging, and Production environments.

FR6 — Feature Evaluation: The system must determine whether a feature should be available to a user by checking the global setting and any applicable user-specific setting.

5. Business Rules

Business rules define the conditions the system must follow when managing features.

BR1 — Application Ownership: Every registered feature must belong to an application registered in the system.

BR2 — Environment Separation: A feature may be enabled in one environment and disabled in another. For example, a feature may be enabled in Development but disabled in Production.

BR3 — Global Switch: If a feature is disabled globally for an environment, it must be unavailable to all users in that environment, regardless of their individual settings.

BR4 — User Targeting: When a feature is globally enabled, administrators may restrict its availability to selected users.

BR5 — Default Availability: If no user-specific restriction has been configured, the feature is available to all users when its global setting is enabled, subject to other eligibility and security checks.

BR6 — Feature Availability: An enabled feature flag does not automatically grant a user permission to perform an action. Normal authentication, authorization, eligibility, and other applicable security checks must still be enforced.

6. Scope of the Project

This project focuses on designing a system for managing feature availability. The betting feature will be used as an example to demonstrate how administrators can control access to a feature.

The project covers the requirements, business rules, database design, and explanation of how the system would work. It does not require the group to build a complete betting service or implement a working application unless the project requirements change.

7. Expected Outcome

The expected outcome is a clear design for a system that allows Tatum Bank to manage application features centrally, control their availability across environments, and selectively make them available to users without requiring a new application release for every setting change.